Lesson 22: Conducting Surveys

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 7.4: Conducting Surveys; 7.5: Survey Research (Summary).

In this lesson

A survey is only as good as its sample. You will learn the main types of sampling, why about 1,000 respondents can represent millions, how sampling bias and non-response bias distort results, and the pros and cons of four ways to conduct a survey. In English, you will practice comparatives and superlatives for comparing methods and the language of approximate numbers.

Learning goals

After this lesson you should be able to:

- Distinguish probability from non-probability sampling and describe simple random, stratified and cluster sampling.
- Explain what a 95% confidence interval shows and why sample size matters more than population size.
- Define sampling bias and non-response bias, and list ways to raise the response rate.
- Compare in-person, telephone, mail and internet surveys.
- Use comparatives, superlatives and approximation words accurately.

Key language

Term or phrase	Meaning	Example
probability sampling	the researcher can specify the probability that each member of the population will be selected	election polls
non-probability sampling	the researcher cannot specify these probabilities; e.g., convenience sampling (people nearby and willing)	most psychological research
sampling frame	a list of all the members of the population, from which the sample is selected	a list of registered voters
simple random sampling	each member of the population has an equal probability of being selected	random-digit dialing
stratified random sampling	the population is divided into subgroups (strata) and a random sample is taken from each	about 126 African Americans among 1,000 adults

Term or phrase	Meaning	Example
cluster sampling	clusters (e.g., towns) are sampled first, then individuals within them	10 towns, 20 residents each
confidence interval	a range computed so that, usually 95% of the time, the population value lies within it	47% to 53%
sampling bias	a sample that is not representative of the population and so produces inaccurate results	the *Literary Digest* poll
non-response bias	sampling bias that occurs when non-responders differ systematically from responders	non-responders who did not drink
by far the most	much more than any other	"in-person interviewing is by far the most costly approach"
in large part	mainly	"Probability sampling was developed in large part to address the issue of sampling bias."

Rules and explanations

The ideas

Two kinds of sampling. Most psychological research uses *non-probability sampling*: convenience, snowball (participants recruit others), quota (subgroups in proportion to the population) or self-selection (people volunteer on their own). Survey researchers prefer *probability sampling*, because they want accurate estimates of what is true in a population (elections are often decided by a few percentage points). It needs (1) a clearly specified population and (2) a *sampling frame*, such as telephone directories, voter lists, hospital or insurance records, or a map.

Probability method	How it works	Book's example
Simple random	every member has an equal chance	random-digit dialing
Proportionate stratified	strata sampled in proportion to the population	12.6% of Americans are African American, so about 126 in 1,000
Disproportionate stratified	extra respondents from small subgroups, so that conclusions about them are valid	Asian Americans (about 5.6%)
Cluster	random clusters, then random individuals within them; no list of every member of the population is needed	10 small towns with 20 residents each, instead of 200 towns

How large a sample? It depends on the confidence wanted and the budget; most surveys use about 100 to 1,000 respondents, and a power analysis guides the trade-off. If exactly 50% of a sample intend to vote for the incumbent, the 95% confidence interval is 40% to 60% with 100 voters, 47% to 53% with 1,000, and 48% to 52% with 2,000. The interval shrinks ever more slowly, so going beyond about 1,000 is often not worth the extra time, effort and money. Surprisingly, it depends on the size of the

sample, not of the population (as long as the population is much larger than the sample). Note: 1,000 out of 252 million adults is about 0.0004%; the book prints "0.000004%," which is the proportion, not the percentage.

Sampling bias. The *Literary Digest* drew its mailing lists from telephone directories and lists of registered automobile owners, which over-represented wealthier people, who were more likely to vote for Landon. Gallup knew about this bias and also sampled less wealthy people.

Non-response bias affects even careful random samples. In a mail survey on alcohol, Lahaut et al. (2002) received answers from only about half the sample after two reminders. Unannounced home visits to some non-responders (up to five times) showed that they included an especially high proportion of abstainers, so estimates based on responders alone were too high. Statistical corrections rest on assumptions that may be wrong, so the best approach is to maximize the *response rate* (the proportion of selected people who respond):

- choose a mode with a high response rate;
- send a short pre-notification and follow-up reminders;
- keep the questionnaire short, simple and on topic;
- offer an incentive, especially cash, but not one so large that it becomes coercive.

Four ways to conduct a survey

Mode	Response rate	Cost	Notes
In person	highest	by far the most costly	closest contact, e.g., when interviewers must judge respondents
Telephone	lower	costly, but less so	directories now miss people with only cell phones
Mail	even lower	less costly still	most susceptible to non-response bias
Internet	low, like mail	low	email links suit organizations such as a university; website links rarely give random samples; likely to become dominant

Gosling et al. (2004) found that common worries about web studies are largely myths: internet samples, for example, are more diverse than traditional ones, though not fully representative. Amazon's Mechanical Turk (over 500,000 workers in over 190 countries) makes online data collection cheap.

Language focus

1. Comparing methods.

- superlative: "In-person interviews have *the highest* response rates and provide *the closest* personal contact with respondents."
- comparative + *still* / *even*: "Mail surveys are *less costly still* but generally have *even lower* response rates."

- *the* + comparative, *the* + comparative: "*The larger* the sample, *the closer* any statistic based on that sample will tend to be to the corresponding value in the population."

2. Approximate numbers. Writers signal how exact a number is: *about*, *roughly* (either side: "the roughly 252 million adults"), *over* (above: "over 500,000 workers"), *less than* (below: "less than 5 minutes"), *up to* (a maximum: "up to five times") and *only* (the number is small: "only about half the sample").

> **Note for Arabic speakers** - Arabic often expresses "the more ..., the more ..." with a "whenever ..., ..." pattern. In English, use *the* + comparative in both clauses: "The larger the sample, the narrower the confidence interval." - Use *percent* after a number ("12.6 percent") and *percentage* without one ("a small percentage of the population").

Practice

1. Reading questions

- 1.1 Why do survey researchers prefer probability sampling?
- 1.2 Why might a researcher use disproportionate stratified random sampling?
- 1.3 With 1,000 respondents, 50% of whom intend to vote for the incumbent, what is the 95% confidence interval? Would it change if the population were ten times larger?
- 1.4 Why was the *Literary Digest* poll of 1936 wrong?
- 1.5 How did Lahaut et al. (2002) test for non-response bias, and what did they find?
- 1.6 Which survey mode is most susceptible to non-response bias, and why?

2. The book's exercises (Section 7.5)

- 2.1 Discussion: Think of a question that each of the following professionals might try to answer using survey research. (a) a social psychologist (b) an educational researcher (c) a market researcher who works for a supermarket chain (d) the mayor of a large city (e) the head of a university police force
- 2.2 Discussion: Write a survey item and then write a short description of how someone might respond to that item based on the cognitive model of survey responding (or choose any item on the Rosenberg Self-Esteem Scale; the book's link no longer works, see "Watch online").
- 2.3 Practice: Write survey items for each of the following general questions. In some cases, a series of items, rather than a single item, might be necessary. (a) How much does the respondent use Facebook? (b) How much exercise does the respondent get? (c) How likely does the respondent think it is that the incumbent will be re-elected in the next presidential election? (d) To what extent does the respondent experience "road rage"?
- 2.4 Discussion: If possible, identify an appropriate sampling frame for each of the following populations. If there is no appropriate sampling frame, explain why. (a) students at a particular university (b) adults living in the state of Washington (c) households in Pullman, Washington (d) people with low self-esteem

- 2.5 Practice: Use one of the online survey creation tools to create a 10-item survey questionnaire on a topic of your choice. (Self-study: build it, but do not send it to respondents.)

3. Language

- 3.1 Complete the book's summary: "In-person interviewing has ____ (high) response rates but is ____ (expensive). Mail and internet surveys are ____ (expensive) but have much ____ (low) response rates."
- 3.2 Rewrite with *the ..., the ...*: "Larger samples provide greater confidence."

Speak with your tutor

- Explain stratified and cluster sampling with examples from your university; ask the tutor to check your comparatives.
- In PSYC 110 Lecture 13 (Why Are People Different?: Differences), Professor Bloom criticizes a phone questionnaire study suggesting that family meals help teenagers avoid drugs (for example, for ignoring the direction of causality). Using Lessons 21-22, discuss two more problems such a survey may face (for example, the "edit" step, non-response).
- In PSYC 110 Lecture 12 (Emotions, Part II), Bloom says that the culture-of-honor researchers chose white, non-Hispanic, non-Jewish male students from a subject pool to get a homogeneous sample. What kind of sampling is this, and to whom can the results be generalized?
- Ask the tutor to play an ethics committee member and defend your survey plan: incentives, confidentiality of online data, non-response bias.

Writing task

Research project: final survey items and procedure (200-300 words, plus your revised items).

1. *Pilot your items*. Ask your tutor how it interprets each Lesson 21 item. Revise at least two, noting in one line what you changed and why.
2. *Write a "Sampling and procedure" paragraph* for your Method section:
    - the population and sampling frame (e.g., the registrar's list of undergraduates at your university);
    - a probability sampling method and why (e.g., proportionate stratified sampling by year of study);
    - the sample size, justified by the trade-off between confidence and cost;
    - the mode (e.g., an emailed link), compared with one alternative;
    - three ways to maximize the response rate, and a check for non-response bias (e.g., following up some non-responders, as Lahaut et al., 2002, did);
    - an ethical point about incentives or online data.

Use the future passive (*will be selected*), a comparative or superlative, and approximation words. If you work on your own topic, define its population and sampling frame in the same way.

Self-editing checklist

- Is the population precisely defined, with a real sampling frame?
- Is the method truly probability sampling?
- Is the sample size justified, not just stated?
- Are the response-rate strategies specific?
- Is the incentive small enough not to be coercive?
- Are sources cited in APA style?
- Do the revised items still meet all five BRUSO criteria?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- The Rosenberg Self-Esteem Scale, for exercise 2.2 (current page; the book's link is dead): https://socy.umd.edu/about-us/using-rosenberg-self-esteem-scale
- PsyToolkit, a free, noncommercial survey tool listed in Section 7.4: https://www.psytoolkit.org/

Answer key

1. Reading questions

- 1.1 Their goal is accurate estimates of what is true in a population, and these are most accurate when based on a probability sample.
- 1.2 To include enough respondents from a small subgroup (e.g., Asian Americans) to draw valid conclusions about it.
- 1.3 47% to 53%. No: confidence intervals depend on the size of the sample, not of the population ("Sample Size and Population Size").
- 1.4 Sampling bias: its lists (telephone directories, registered automobile owners) over-represented wealthier people, who were more likely to vote for Landon.
- 1.5 Unannounced home visits, up to five times, to a subset of non-responders. The non-responders included an especially high proportion of abstainers, so the original estimates were too high.
- 1.6 Mail surveys, because they generally have even lower response rates than telephone surveys.

2. The book's exercises (example answers; many others are possible)

- 2.1 (a) What percentage of adults hold negative stereotypes about older people? (b) Is students' weekly study time related to their grades? (c) Which products would customers like the stores to sell? (d) How satisfied are residents with public transport? (e) How safe do students feel on campus at night?
- 2.2 "During the past month, how often have you felt stressed about your coursework?" (Never to Always). *Interpret*: does "stressed" mean any worry? *Retrieve*: recall stressful days, or a belief such as "I am a calm person." *Judgment*: "most weeks." *Format*: *Often* or *Sometimes*? *Edit*: choose *Sometimes* to avoid looking unable to cope.
- 2.3 (a) "In a typical week, on how many days do you use Facebook? ____ days," then "On those days, about how much time do you spend on it?" (Less than 30 minutes / 30 to 59 minutes / 1 hour to less than 2 hours / 2 hours or more). (b) "In a typical week, on how many days do you exercise for at least 30 minutes? ____ days." (c) "How likely do you think it is that the current president will be re-elected in the next presidential election?" (Extremely Unlikely / Somewhat Unlikely / As Likely as Not / Somewhat Likely / Extremely Likely). (d) A series of frequency items about specific behaviors instead of the vague label: "In a typical month, how often do you shout at another driver?" and "... make an angry gesture at another driver?" (Never / Rarely / Sometimes / Often / Always).
- 2.4 (a) Yes: the university's list of enrolled students. (b) No complete list exists (voter lists and telephone directories miss many adults); a map can serve as a frame for cluster sampling. (c) Yes: a map or address list of the town, from which households can be selected. (d) No: there is no list of people with low self-esteem, a trait that must first be measured; researchers would have to sample a broader population and screen it.
- 2.5 Check it against the self-editing checklists of Lessons 21 and 22.

3. Language

- 3.1 the highest; the most expensive; less expensive; lower.
- 3.2 The larger the sample, the greater the confidence.

My notes